
Problem Set 3

Problem 2.24

A stream of warm water is produced in a steady-flow mixing process by combining 1.0 kg s^{-1} of cool water at 25 degC with 0.8 kg s^{-1} of hot water at 75 deg C . During mixing, heat is lost to the surroundings at a rate of 30 kJ s^{-1} . What is the temperature of the warm water stream? Assume the specific heat of water is constant at $4.18 \text{ kJ kg}^{-1} \text{ K}^{-1}$.

Problem 2.28

Nitrogen flows at steady state through a horizontal, insulated pipe with inside diameter of 1.5 (in). A pressure drop results from flow through a partially opened valve. Just upstream from the valve the pressure is 100 (psia), the temperature is 120 (degF), and the average velocity is 20 (ft)(s)^{-1} . If the pressure just downstream from the valve is 20 (psia), what is the temperature? Assume for nitrogen that PV/T is constant, $C_v = (5/2)R$, and $C_p = (7/2)R$. (Values of R are given in App. A.)

Problem 2.40

One kilogram of air is heated reversibly at constant pressure from an initial state of 300 K and 1 bar until its volume triples. Calculate W , Q , ΔU , and ΔH for the process. Assume for air that $(PV/T) = 83.14 \text{ (bar)(cm)}^3 \text{ (mol)}^{-1} \text{ (K)}^{-1}$ and $C_p = 29 \text{ (J)(mol)}^{-1} \text{ (K)}^{-1}$. Report your answers in kJ.

Problem 2.38

Carbon dioxide gas enters a water-cooled compressor at conditions $P_1 = 15 \text{ (psia)}$ and $T_1 = 50 \text{ (degF)}$, and is discharged at conditions $P_2 = 520 \text{ (psia)}$ and $T_2 = 200 \text{ (degF)}$. The entering CO_2 flows through a 4-inch-diameter pipe with a velocity of 20 (ft)(s)^{-1} , and is discharged through a 1-inch-diameter pipe. The shaft work supplied to the compressor is $5,360 \text{ (Btu)(lb mol)}^{-1}$. What is the heat-transfer rate from the compressor in (Btu)(hr)^{-1} ?

Additional Information:

$$H_1 = 307 \text{ (Btu)(lb}_m\text{)}^{-1} \text{ and } V_1 = 9.25 \text{ (ft)}^3 \text{ (lb}_m\text{)}^{-1}$$

$$H_2 = 330 \text{ (Btu)(lb}_m\text{)}^{-1} \text{ and } V_2 = 0.28 \text{ (ft)}^3 \text{ (lb}_m\text{)}^{-1}$$